

SMQTT (Secure MQTT)

SMQTT is an enhanced version of MQTT that prioritizes security for IoT applications, particularly for resource-constrained devices. It integrates advanced cryptographic mechanisms, such as attribute-based encryption (ABE), to ensure secure, fine-grained access control.

Features

1. Attribute-Based Encryption (ABE): Encrypts messages based on attributes (e.g., device role, location), allowing only authorized subscribers to decrypt.
2. Lightweight Cryptography: Uses efficient algorithms (e.g., elliptic curve cryptography, ECC) to minimize computational overhead on low-power devices (e.g., 8-bit MCUs).
3. End-to-End Security: Messages remain encrypted through the broker, ensuring confidentiality.
4. Policy-Based Access: Defines access policies (e.g., “only managers in building A”) enforced by ABE.
5. Compatibility: Retains MQTT’s publish-subscribe model, integrating with existing brokers.
6. Scalability: Supports secure communication in large IoT networks with thousands of devices.
7. Low Overhead: Adds minimal latency compared to standard MQTT, suitable for constrained networks.

Working Principle

- SMQTT builds on MQTT by adding a security layer:
 1. Key Distribution: A trusted key management authority generates public/private key pairs based on attributes (e.g., role=admin, location=room1).
 2. Message Encryption: The publisher encrypts the message with an ABE policy (e.g., “accessible by devices with role=admin”).
 3. Broker Routing: The broker forwards the encrypted message to subscribers without decrypting it, preserving MQTT’s topic-based routing.
 4. Decryption: Subscribers with matching attributes (e.g., role=admin) use their private keys to decrypt the message.
 5. Authentication: Optional client authentication (e.g., via certificates) ensures only authorized devices connect.
- Example:
 - o A sensor in a smart factory publishes encrypted data to secure /factory/temperature with a policy “accessible by maintenance team in sector B.”
 - o Only devices with attributes role=maintenance and sector=B can decrypt the message, even if others subscribe to the topic.

Applications

1. Smart Grids: Securely transmit energy consumption data to authorized control systems.

2. Healthcare: Protect sensitive patient data (e.g., heart rate) from unauthorized access.
3. Industrial IoT: Secure machine-to-machine communication in factories with role based access.
4. Smart Homes: Restrict access to critical controls (e.g., smart locks) to family members.
5. Defence IoT: Secure sensor data in military applications with strict access policies.

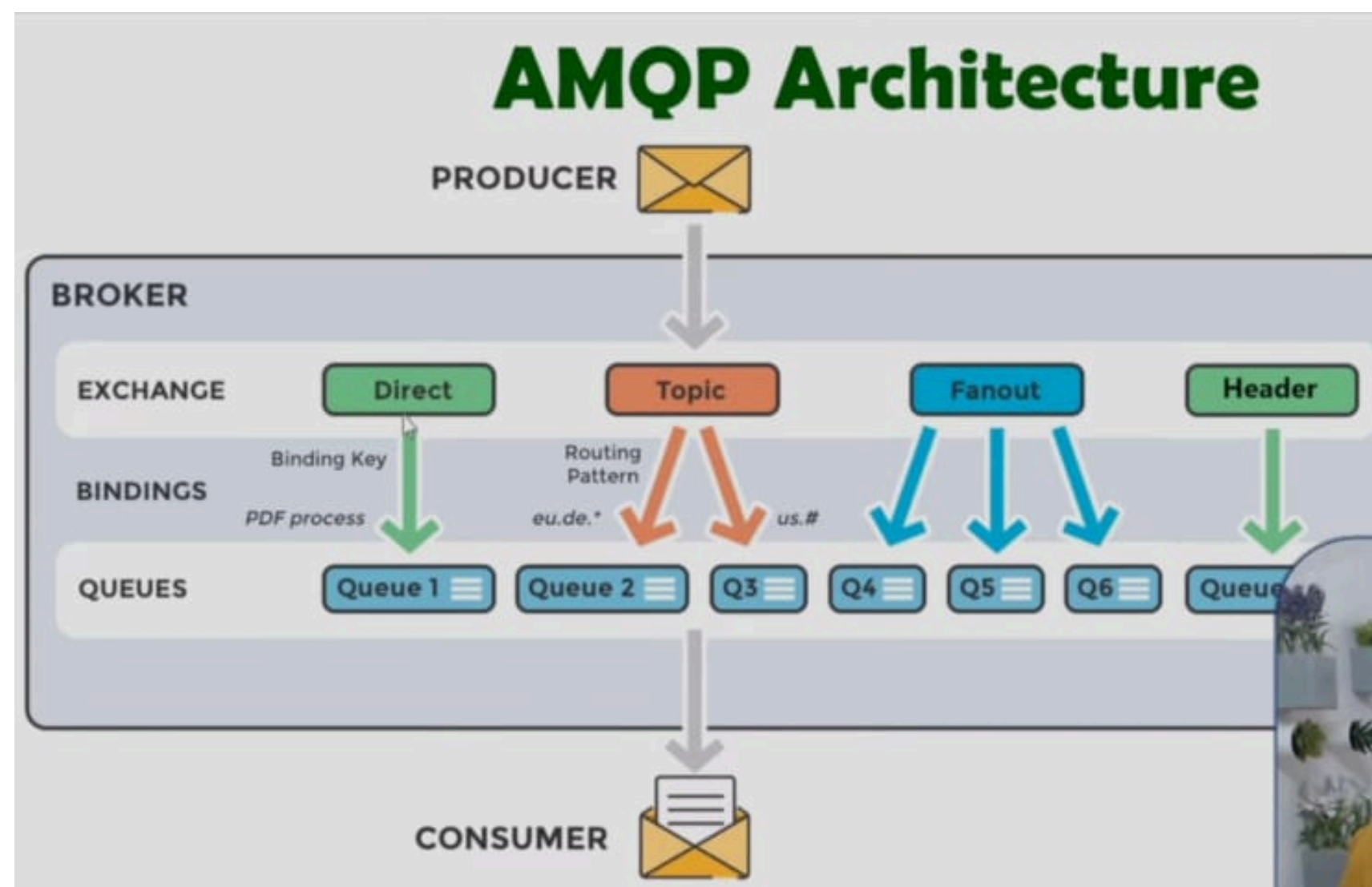
AMQP (Advanced Message Queuing Protocol)

AMQP is a robust, open-standard protocol for message-oriented middleware, designed for reliable, asynchronous message queuing in enterprise and IoT applications. It supports complex routing and guaranteed delivery, making it suitable for high-throughput systems.

Basics of AMQP

- AMQP-Advanced Message Queuing Protocol
- It is an open standard application layer protocol for message-oriented middleware.
- AMQP is mainly used in transaction messages between servers and users.
- Mostly it is used in banking servers and banking applications.
- It uses TCP for reliable communication.
- End points acknowledge acceptance of each message.
- It follows Public Subscriber model for communication.

Architecture of AMQP



The image shows the **AMQP Architecture** (Advanced Message Queuing Protocol), which is a standard protocol used in **message-oriented middleware**. It enables systems to communicate by sending messages through a **broker** system, decoupling the sender (producer) and receiver (consumer).

1. Producer

- The **producer** is the sender of messages. It sends messages to the **broker** (not directly to the consumer).
- The producer does **not need to know** who will consume the message.

2. Broker

- The **broker** is the middleware that routes messages from producers to consumers using **exchanges**, **bindings**, and **queues**.

3. Exchange

- An **exchange** is responsible for routing messages to the correct **queue(s)** based on specific rules. There are different types:
 - **Direct**: Routes messages with a specific binding key to the matching queue.
 - **Topic**: Routes messages based on a **pattern-matching** routing key.
 - **Fanout**: Broadcasts the message to **all bound queues**, ignoring routing keys.
 - **Header**: Routes messages based on **message headers**, not routing keys.

4. Bindings

- A **binding** is the link between an exchange and a queue.
- It uses **routing logic**, such as:
 - o Binding Key: e.g., "PDF process" for direct exchange.
 - o Routing Pattern: e.g., "eu.de.*" or "us.#" for topic exchange.

5. Queues

- **Queues** hold messages until they are consumed.
- Each queue can be linked to one or more exchanges with specific binding rules.
- In the diagram:
 - o **Queue 1** gets messages from the **Direct** exchange.
 - o **Queue 2, Q3** get messages from the **Topic** exchange based on different routing patterns.
 - o **Q4, Q5, Q6** get messages from **Fanout** (all of them get the same message).
 - o **Queue 7** is connected to **Header** exchange (based on header content).

3. Consumer

- The **consumer** is the receiver of messages.
- It reads messages from the queue.
- Like the producer, it doesn't need to know who sent the message.

Features

1. **Reliable Delivery**: Ensures messages are delivered with acknowledgments and transactions.
2. **Flexible Routing**: Supports multiple exchange types (direct, topic, fanout, headers) for precise message distribution.
3. **Asynchronous Messaging**: Decouples producers and consumers, enabling high scalability.
4. **Security**: Uses TLS for encryption, SASL for authentication, and access control (e.g., virtual hosts in RabbitMQ).

5. Message Durability: Persists messages in queues until consumed, surviving broker restarts.
6. Cross-Platform: Interoperable across languages (e.g., Python, Java) and brokers (e.g., RabbitMQ, Apache Qpid).
7. High Throughput: Handles thousands of messages per second in distributed systems.
8. Transactional Support: Groups messages into atomic transactions for consistency.
9. Message Prioritization: Allows high-priority messages (e.g., alerts) to be processed first.

Components

1. Producer: Sends messages to an exchange (e.g., IoT sensor publishing temperature).
2. Consumer: Retrieves messages from a queue (e.g., analytics server processing data).
3. Exchange: Routes messages to queues based on routing rules:
 - o Direct: Matches exact routing key (e.g., sensor.temperature).
 - o Topic: Matches patterns (e.g., sensor.*.temp for all rooms).
 - o Fanout: Broadcasts to all bound queues (e.g., alerts to multiple systems).
 - o Headers: Routes based on message header attributes (e.g., priority=high).
4. Queue: Stores messages until consumed, with options for durability and expiration.
5. Broker: Manages exchanges, queues, and bindings (e.g., RabbitMQ, ActiveMQ).
6. Binding: Links exchanges to queues with routing keys or conditions.
7. Virtual Host: Logical isolation within a broker for multi-tenant setups.

AMQP Frames



- Frame header: The header has a fixed size (8 bytes), and it is mandatory, as it contains the information needed to parse the rest of the frame itself—for example the total frame size and the frame type.
- Extended header: A variable header that depends on the frame type.
- Frame body: A sequence of bytes that has a format that depends on the frame type.
- Frame types can be Open, Close, begin, end, attach, detach, transfer, flow and disposition.

Communication

- Model: Producer–exchange–queue–consumer, with asynchronous message queuing:
 1. Connection: Producer/consumer connects to the broker, sending a protocol header and negotiating parameters (e.g., heartbeat interval).
 2. Channel Setup: Opens a logical channel for communication (method frame: channel. Open).
 3. Exchange/Queue Setup: Declares exchanges and queues (e.g., exchange. declare, queue. declare) and binds them (e.g., queue. bind with routing key sensors.*).
 4. Publishing: Producer sends a message via method frame (basic. publish), content header (properties), and body frame(s) to an exchange.
 5. Routing: The exchange routes the message to queues based on bindings (e.g., topic exchange matches sensors.room1.temp to sensors.*.temp).
 6. Consumption: Consumer retrieves messages from a queue (e.g., basic. consume), processes them, and sends acknowledgments (basic ack) for reliability.
 7. Durability: Persistent queues/messages survive broker restarts; transactions (tx. select, tx. commit) ensure atomicity.
- Example:
 - o A sensor (producer) publishes a message to a topic exchange with routing key sensors.room1.temp and payload {"temp": 25.5}.
 - o The broker routes it to two queues bound to sensors.*.temp (e.g., for analytics and alerts).
 - o An analytics app (consumer) retrieves the message, processes it, and sends an ACK to the broker.

Applications

1. Industrial IoT: Queues sensor data for real–time analytics or maintenance alerts.
2. Smart Cities: Routes traffic or environmental data to control systems.
3. Logistics: Tracks shipments with reliable, high–throughput messaging.
4. Financial Systems: Processes stock trades or payment transactions with guaranteed delivery.
5. Smart Grids: Coordinates energy devices with asynchronous, durable messaging.
6. Micro services: Integrates IoT with enterprise systems via message queues.

Introduction of IEEE 802.15.4 Technology

IEEE 802.15.4 is a low–cost, low–data–rate wireless access technology for devices that are operated or work on batteries. This describes how low–rate wireless personal area networks (LR–WPANs) function.

IEEE 802.15.4e:

802.15.4e for industrial applications and 802.15.4g for the smart utility networks (SUN)

The 802.15.4e improves the old standard by introducing mechanisms such as time slotted access, multichannel communication and channel hopping.

- **IEEE 802.15.4e introduces the following general functional enhancements:**

1. Low Energy (LE): This mechanism is intended for applications that can trade latency for energy efficiency. It allows a node to operate with a very low duty cycle.
2. Information Elements (IE) It is an extensible mechanism to exchange information at the MAC sublayer.
3. Enhanced Beacons (EB): Enhanced Beacons are an extension of the 802.15.4 beacon frames and provide a greater flexibility. They allow to create application-specific frames.
4. Multipurpose Frame: This mechanism provides a flexible frame format that can address a number of MAC operations. It is based on IEs.
5. MAC Performance Metric: It is a mechanism to provide appropriate feedback on the channel quality to the networking and upper layers, so that appropriate decision can be taken.
6. Fast Association (FastA) The 802.15.4 association procedure introduces a significant delay in order to save energy. For time-critical application latency has priority over energy efficiency.

- **IEEE 802.15.4e defines five new MAC behavior modes.**

1. Time Slotted Channel Hopping (TSCH): It targets application domains such as industrial automation and process control, providing support for multi-hop and multichannel communications, through a TDMA approach.
2. Deterministic and Synchronous Multi-channel Extension (DSME): It is aimed to support both industrial and commercial applications.
3. Low Latency Deterministic Network (LLDN): Designed for single-hop and single channel networks
4. Radio Frequency Identification Blink (BLINK): It is intended for application domains such as item/people identification, location and tracking.
5. Asynchronous multi-channel adaptation (AMCA): It is targeted to application domains where large deployments are required, such as smart utility networks, infrastructure monitoring networks, and process control networks.

Properties:

1. Standardization and alliances: It specifies low-data-rate PHY and MAC layer requirements for wireless personal area networks (WPAN).

IEEE 802.15. Protocol Stacks include:

- **ZigBee:** ZigBee is a Personal Area Network task group with a low rate task group 4. It is a technology of home networking. ZigBee is a technological standard created for controlling and sensing the network. As we know that ZigBee is the Personal Area network of task group 4 so it is based on IEEE 802.15.4 and is created by Zigbee Alliance.
- **6LoWPAN:** The 6LoWPAN system is used for a variety of applications including wireless sensor networks. This form of wireless sensor network sends data as packets and uses IPv6 – providing the basis for the name – IPv6 over Low power Wireless Personal Area Networks.
- **ZigBee IP:** Zigbee is a standards-based wireless technology that was developed for low-cost and low-power wireless machine-to-machine (M2M) and internet of things (IoT) networks.
- **ISA100.11a:** It is a mesh network that provides secure wireless communication to process control.
- **Wireless HART:** It is also a wireless sensor network technology, that makes use of time-synchronized and self-organizing architecture.
- **Thread:** Thread is an IPv6-based networking protocol for low-power Internet of Things devices in IEEE 802.15.4-2006 wireless mesh network. Thread is independent.

2. Physical Layer: This standard enables a wide range of PHY options in ISM bands, ranging from 2.4 GHz to sub-GHz frequencies. IEEE 802.15.4 enables data transmission speeds of 20 kilobits per second, 40 kilobits per second, 100 kilobits per second, and 250 kilobits per second. The fundamental structure assumes a 10-meter range and a data rate of 250 kilobits per second. To further reduce power usage, even lower data rates are possible. IEEE 802.15.4 regulates the RF transceiver and channel selection, and even some energy and signal management features, at the physical layer. Based on the frequency range and data performance needed, there are now six PHYs specified. Four of them employ frequency hopping techniques known as Direct Sequence Spread Spectrum (DSSS). Both PHY data service and management service share a single packet structure so that they can maintain a common simple interface with MAC.

3. MAC layer: The MAC layer provides links to the PHY channel by determining that devices in the same region will share the assigned frequencies. The scheduling and routing of data packets are also managed at this layer. The 802.15.4 MAC layer is responsible for a number of functions like:

- Beacons for devices that operate as controllers in a network.
- used to associate and dissociate PANs with the help of devices.
- The safety of the device.
- Consistent communication between two MAC devices that are in a peer-to-peer relationship.

Several established frame types are used by the MAC layer to accomplish these functions. In 802.15.4, there are four different types of MAC frames:

- frame of data
- Frame for a beacon
- Frame of acknowledgement
- Frame for MAC commands

4. Topology: Networks based on IEEE 802.15.4 can be developed in a star, peer-to-peer, or mesh topology. Mesh networks connect a large number of nodes. This enables nodes that would otherwise be out of range to interact with each other to use intermediate nodes to relay data.

5. Security: For data security, the IEEE 802.15.4 standard employs the Advanced Encryption Standard (AES) with a 128-bit key length as the basic encryption technique. Activating such security measures for 802.15.4 significantly alters the frame format and uses a few of the payloads. The very first phase in activating AES encryption is to use the Security Enabled field in the Frame Control part of the 802.15.4 header. For safety, this field is a single bit which is assigned to 1. When this bit is set, by taking certain bytes from its Payload field, a field known as the Auxiliary Security Header is formed following the Source Address field.

6. Competitive Technologies: The IEEE 802.15.4 PHY and MAC layers serve as a basis for a variety of networking profiles that operate in different IoT access scenarios. DASH7 is a competing radio technology with distinct PHY and MAC layers.

The architecture of LR-WPAN Device:

This diagram represents the **IEEE 802.11 (Wi-Fi) protocol stack** architecture, showing the layered structure and how data flows through different layers in wireless communication.

1. Upper Layers

- These layers represent the higher-level network protocols and applications, such as IP, TCP, HTTP, etc.
- They generate data to be transmitted or process data received from lower layers.

2. 802.2 Logic Link Control (LLC)

- This layer is part of the Data Link Layer (Layer 2 in OSI model).
- It provides **logical link control**, managing communication between the upper layers and lower sublayers.
- It handles framing, flow control, and error checking.

3. Service Specific Convergence Sub Layer (SSCS)

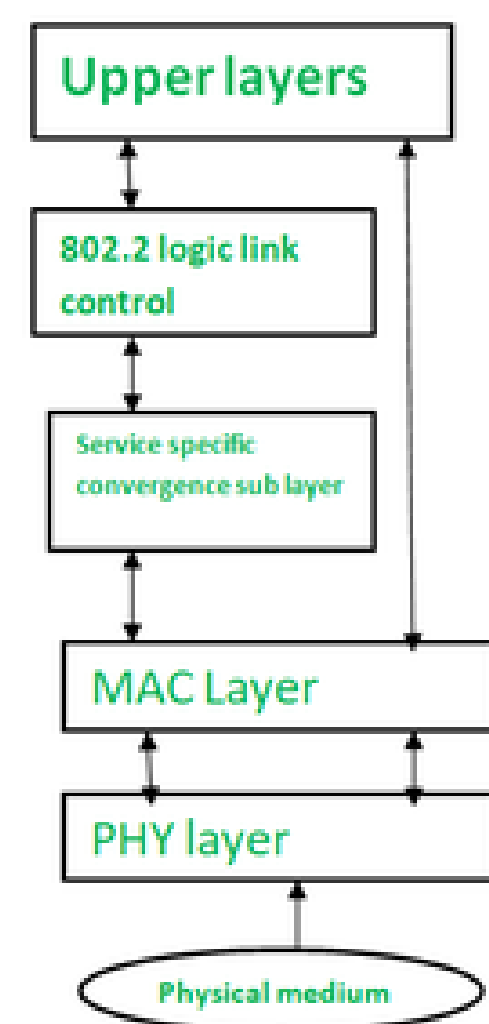
- This sublayer adapts services to meet specific requirements of various network types (like Ethernet, Token Ring, or wireless).
- It acts as an interface between the LLC and the MAC layer, adapting the protocol to different physical networks.

4. MAC Layer (Medium Access Control)

- Responsible for **controlling how devices access the physical medium** (e.g., wireless radio waves).
- Manages addressing, channel access, collision avoidance, and frame formatting.
- In wireless networks, it handles important functions like retransmissions and acknowledgments.

5. PHY Layer (Physical Layer)

- This is the lowest layer in this stack.
- Responsible for **transmitting raw bits over the physical medium** (e.g., radio waves, cables).
- It defines electrical, mechanical, and procedural specifications for the physical connection.



6. Physical Medium

- The actual medium through which data is transmitted (e.g., air for Wi-Fi, fiber optics, copper cables).

Near Field Communication (NFC)

Near Field Communication (NFC) is a short-range wireless communication technology that has revolutionized how devices interact in various industries, from payments to data sharing.

This NFC tutorial guide covers the protocol stack, working principles, frame structure, and different communication modes.

What is NFC (Near Field Communication)?

NFC stands for Near Field Communication. It's a short-range, low data rate wireless communication technology operating at 13.56 MHz frequency. It's a contactless mode of communication using electromagnetic waves.

This technology allows two devices housing an NFC chip to communicate for various applications, for example:

- Data communication between smartphones
- Verification of an authenticated person having an NFC ID card at the office by an NFC reader in the door.
- Banking payments using NFC compliant credit cards to enhance security
- Ticket booking at airports
- Automatic running of scheduled features in an NFC phone.

The following table summarizes features of wireless NFC (Near Field Communication) technology.

How NFC Works

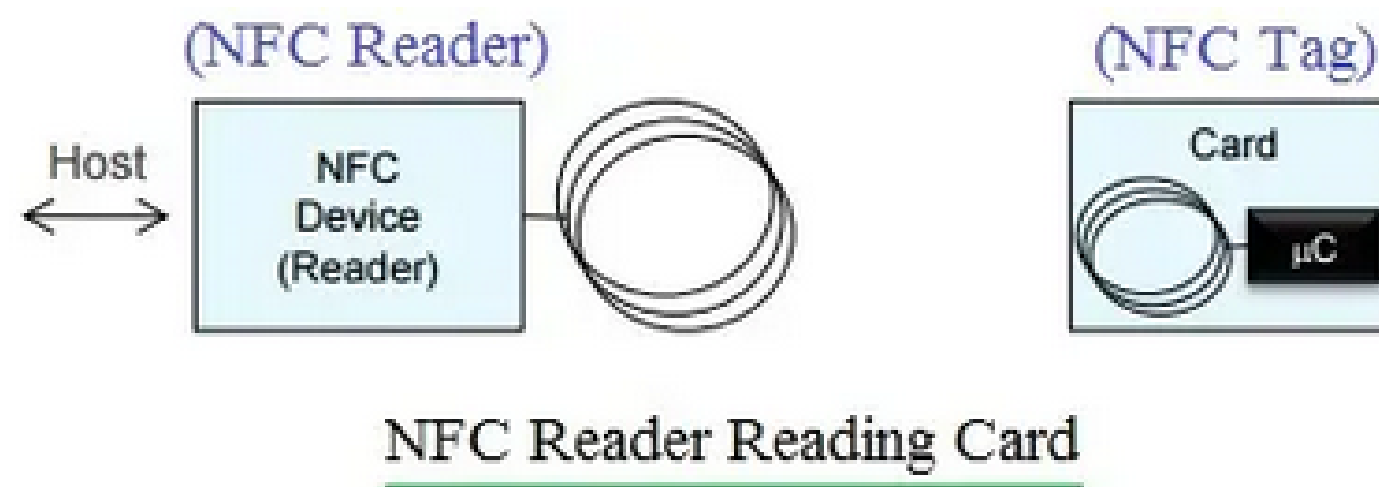


Fig:1 NFC Network

As mentioned, an NFC network consists of two devices known as the initiator device and the target device.

An NFC tag can be an active or passive device. An NFC reader is always an active device. These devices operate in either active-active or active-passive modes.

In active-active mode, both NFC devices have their own power. In active-passive mode, the passive device derives its power from the received EM waves of the active device.

The basic mode of communication in NFC is half-duplex, where one NFC device transmits while the other receives. This is also referred to as "Listen before Talk."

One of the two NFC devices will function as an initiator, which first listens on the channel and transmits only when no other signal is present. The Initiator polls other devices that come closer, and the other NFC device, referred to as the target, listens and responds to the initiator as per the requested message.

NFC Network Modes: Card Emulation, Reader/Writer, Point to Point (Peer)

There are three modes in which NFC tags and readers work: Card Emulation, Reader/Writer, and Peer to Peer (or Point to Point), as explained below.

NFC Card Emulation Mode

In this mode, usually an active device reads a passive device.

EXAMPLE #1: ID cards, NFC tags, or NFC-compliant tickets are read by NFC-compliant active readers at offices or railway stations.

EXAMPLE #2: A smartphone acts as a

Smart card to allow booking tickets or performing online banking transactions or payments using a credit card reader.

Following are the steps involved in card emulation mode:

- Step 1: The NFC reader oscillates at a 13.56 MHz RF field. When the card comes near the RF field, it gets power due to EM (Electro-Magnetic) coupling and connects with the reader.
- Step 2: The reader sends commands using the RF field.
- Step 3: The card responds to the reader as requested.

NFC Reader/Writer Mode

In this mode, one device will be in reading mode, and the other will be in writing mode. They can be either active or passive devices.

- Example #1: Product Authentication
- Example #2: Smart advertising
- Example #3: Device pairing

Following are the steps involved in reader/writer mode:

- Step 1: In read mode, the NFC reader reads NFC tags, either passive or active.
- Step 2: In write mode, one NFC device writes to the other NFC device.

NFC Peer to Peer (Point to Point)

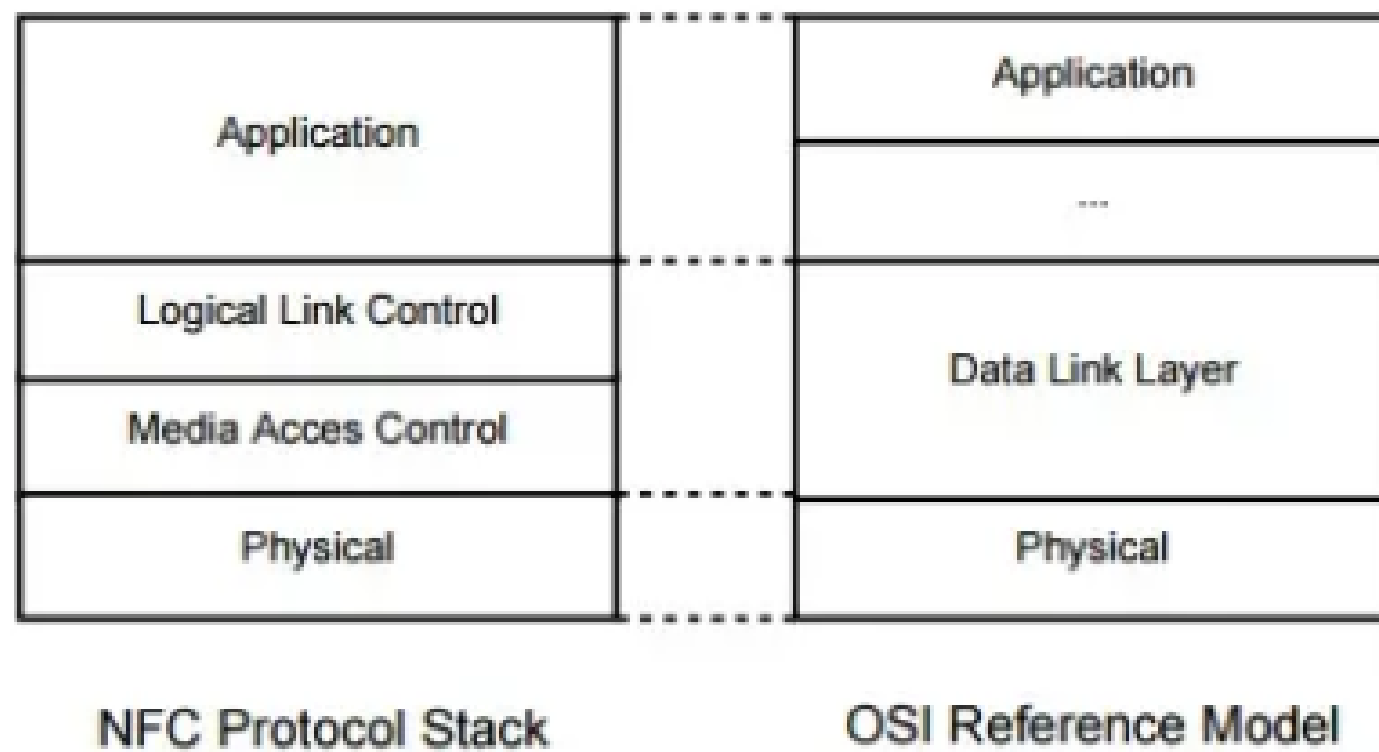
In this mode, data is transmitted from one NFC device to another in ad-hoc or peer-to-peer mode. Any device can become the initiator, and the rest will act as targets to complete the connection establishment.

- Example #1: Smartphone to smartphone communication
- Example #2: Automotive (in-car operation)
- Example #3: Social media applications or games

Following are the steps involved in peer-to-peer mode (active mode):

- Step 1: Both NFC smartphones establish pairing for data communication. Both have power and their own RF fields.
- Step 2: The initiator sends commands by modulating the RF field and switches off the RF field.
- Step 3: The target device responds to the initiator using its own RF field modulation.

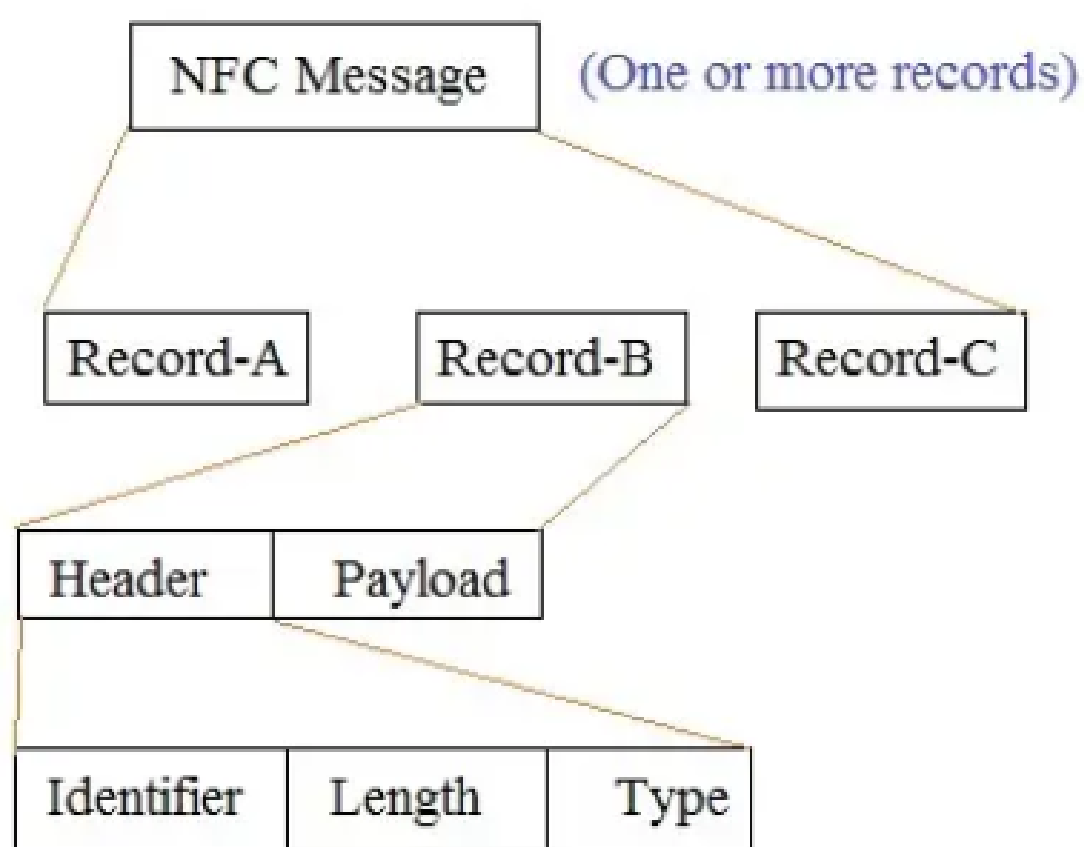
NFC Protocol Stack



The figure depicts the protocol stack of NFC in peer-to-peer mode. As shown, it consists of the application layer, data link layers (logical link control and media access control), and the physical layer.

- **Application Layer:** This layer takes care of the format of the data to be exchanged between NFC devices or between the NFC device and tags. NFC uses NDEF (NFC Data Exchange Format).
- **Data Link Layer:** This layer takes care of different modes of operation and anti-collision mechanisms. LLCP establishes communication between two peer devices.
- **Physical Layer:** This layer takes care of modulation, coding, and RF-related parameters such as frequency and power.

NFC Frame Structure



The NFC frame uses a special encapsulation format referred to as NDEF (NFC Data Exchange Format). Each NFC device transmission is known as a message. One message is composed of one or more records. Each record consists of a header part and a payload part. The header part is made of identifier, length, and type fields. The payload carries data or a URL.